



# **Tamper-Proof Digital Certificate Verification Using Blockchain and QR Authentication**

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# Introduction



- Digital certificates are widely used to verify academic and professional achievements.
- However, traditional paper and PDF certificates can be forged, duplicated, or modified, making verification difficult.
- This project develops a **Java-based certificate verification system** that uses **SHA-256 hashing, QR code authentication, and blockchain concepts** to ensure secure certificate generation and instant verification.
- The system provides a simple, reliable, and tamper-resistant solution for educational institutions and organizations.



# Problem Statement & Objectives



## Problem Statement

- Fake certificates are increasing.
- Manual verification is slow.
- Certificates can be easily modified.
- No reliable verification mechanism.

## Objectives

- Prevent certificate forgery.
- Secure certificate records using blockchain.
- Enable instant QR code verification.
- Detect certificate tampering.
- Maintain transparent audit records.



# Module 1 – Certificate Generation



## Description

This module allows the administrator to create digital certificates using student information. The application generates a unique SHA-256 hash and embeds a QR code into the certificate.

## Process

- Enter student details
- Generate certificate
- Create SHA-256 hash
- Generate QR code
- Save certificate as PDF

## Output

- Digital certificate
- QR-enabled certificate



# Module 2 – Certificate Verification



## Description

Users scan the QR code to verify certificate authenticity. The Java application compares the stored hash with the current certificate hash.

## Process

- Scan QR code
- Retrieve certificate data
- Compare SHA-256 hashes
- Validate certificate
- Display verification status

## Output

- Certificate Verified
- Certificate Invalid
- Verification details



# Module 3 – Admin & Audit Module



## Description

The administrator manages certificate records and monitors verification history. The system records every verification attempt for auditing.

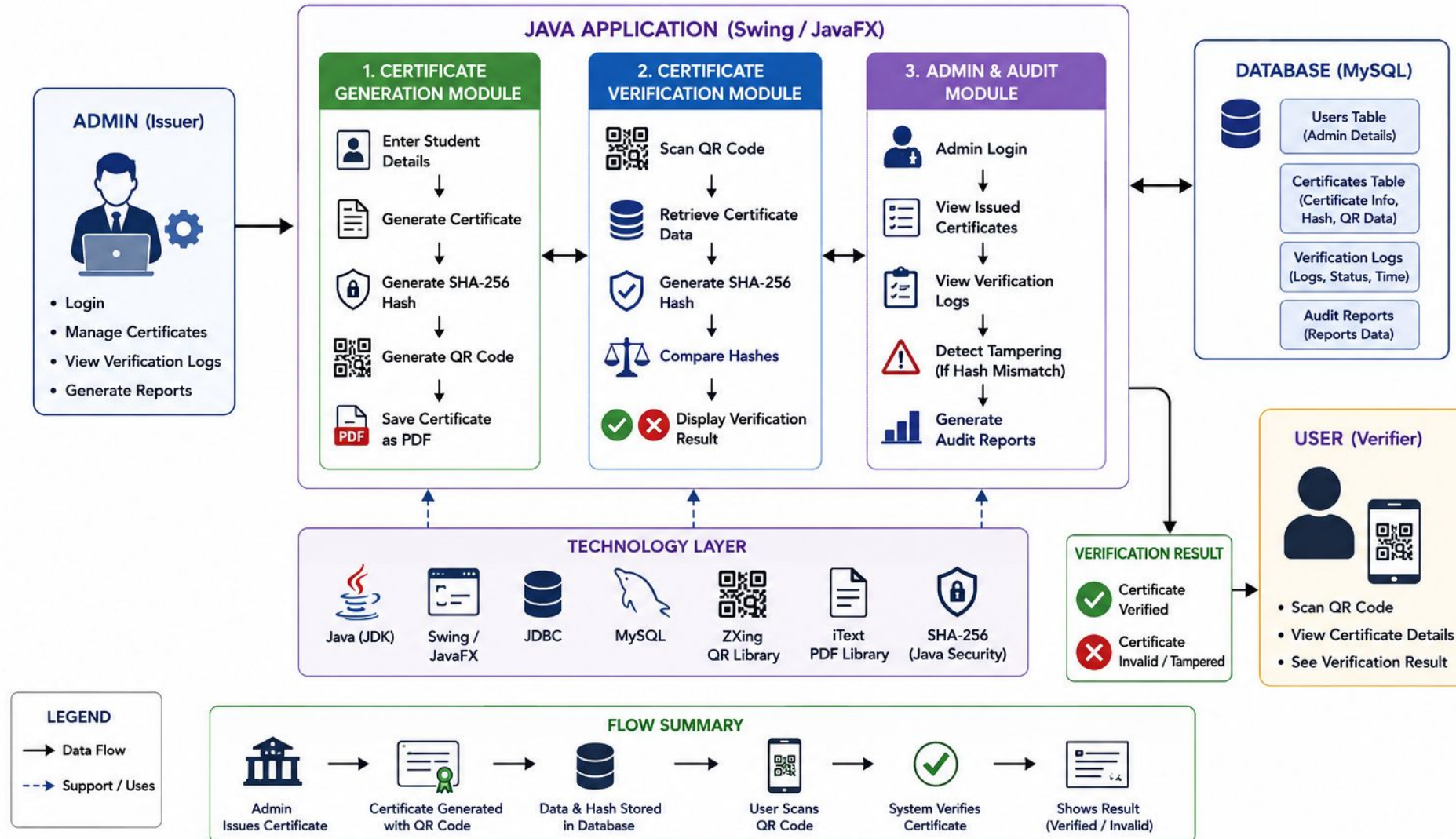
## Process

- Login as administrator
- View issued certificates
- Monitor verification logs
- Detect modifications
- Generate audit reports

## Output

- Audit logs
- Verification history
- Tampering alerts

# System Architecture





# Technologies Used & Applications

## Technologies Used

- Java (JDK 17/21)
- Java Swing / JavaFX
- MySQL
- JDBC
- SHA-256 (Java Security API)
- ZXing QR Code Library
- iText PDF Library
- Eclipse / IntelliJ IDEA

## Applications

- Colleges
- Universities
- Schools
- Corporate Training
- Online Certification
- Government Documents





# Conclusion



- The proposed Java application provides a secure and efficient method for generating and verifying digital certificates.
- By combining **Java, MySQL, JDBC, SHA-256 hashing, and QR code technology**, the system minimizes certificate forgery and simplifies the verification process.
- It offers faster authentication, improved security, and easy certificate management.
- The project demonstrates the practical application of Java programming in solving real-world security challenges and can be further enhanced with cloud integration and mobile-based verification.



# References (2021–2025)



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# THANK YOU!!